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Closing the existence problem for geometric (n_4) configurations

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Abstract Text:

A geometric (n_4) configuration is a set of n points and n lines in the real projective plane such that every point lies on exactly four of the lines and every line contains exactly four of the points. The question of which n admit such a configuration was posed by Grünbaum in 2000, following his 1990 paper with Rigby that gave the first geometric 4-configuration, a (21_4) . Successive work of Bokowski–Schewe, Bokowski–Grünbaum–Schewe, Bokowski–Pilaud and Cuntz settled every case but one. Cuntz’s 2018 paper, which constructed (22_4) and (26_4) configurations, states that only $n = 23$ remained open.

We settle this last case by exhibiting two geometric (23_4) configurations. The first has integer homogeneous coordinates, so that verifying it is an exercise in integer arithmetic. The second is self-polar with respect to a conic and is defined over $\mathbb{Q}(\sqrt{17})$, but not over $\mathbb{Q}$. Consequently geometric (n_4) configurations exist if and only if $n \geq 18$ and $n \neq 19$.

We also explain why $n = 23$ resisted the symmetric constructions that settled the other values. We prove that the group of projective symmetries of any geometric (23_4) configuration has order at most four. Reducing the case of a Klein four-group of symmetries to a few thousand bilinear polynomial systems in 18 unknowns, and solving all of them exactly with Gröbner bases and saturation, shows that our two configurations are the only Klein-symmetric ones, over $\mathbb{R}$ and over $\mathbb{C}$. The same method applied to $n = 22$ yields new (22_4) configurations.

Session Selection:

AMS Special Session on Applied and Computational Algebraic Geometry, I

Subject Level 1:

52 Convex and discrete geometry

Subject Level 2:

52C - Discrete geometry

Subject Level 3:

52C30 - Planar arrangements of lines and pseudolines (aspects of discrete geometry)

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